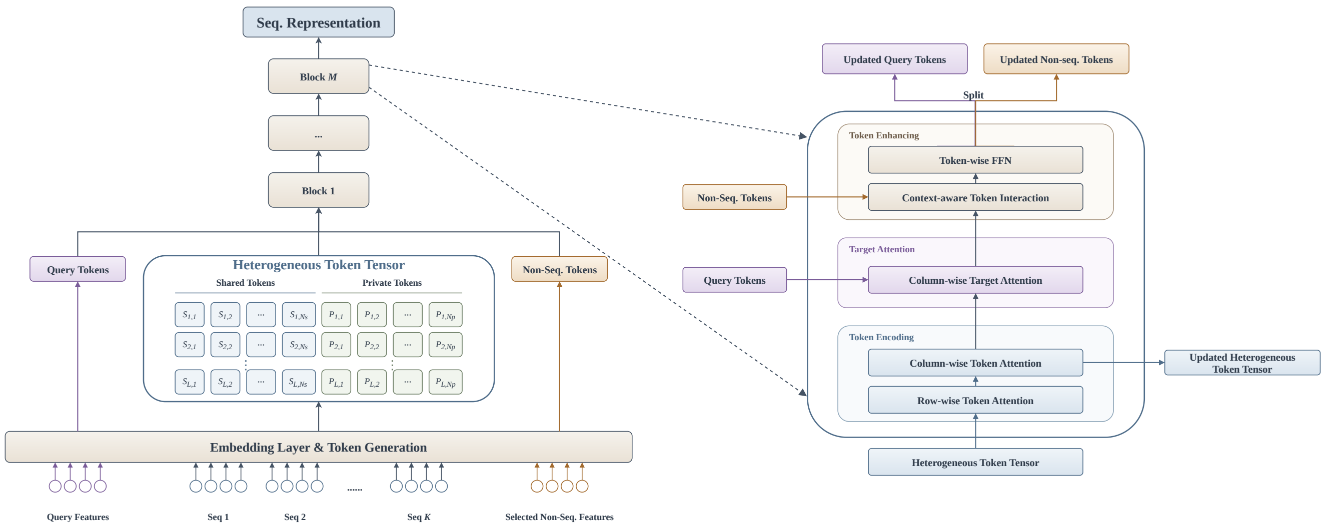

Figure 1: Overview of the proposed SPIRE architecture. Behaviors from heterogeneous sequences are first transformed into shared and private tokens. Building on this decomposition, SPIRE block performs structured token interactions through three modules: (a) Token Encoding for modeling intra-behavior and inter-behavior dependencies; (b) Target Attention for retrieving candidate-relevant interests; and (c) Token Enhancing for incorporating non-sequential information to refine target-aware representations. Multiple SPIRE blocks can be stacked to progressively refine the token representations.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item sets, respectively. For a user $u \in \mathcal{U}$, let $\mathcal{S}_u = \{\mathcal{S}_u^{(1)}, \dots, \mathcal{S}_u^{(K)}\}$ denote the heterogeneous behavior sequences from K scenarios, where $\mathcal{S}_u^{(k)} = (b_{u,t}^{(k)})_{t=1}^{L_u^{(k)}}$ is the k-th sequence with length $L_u^{(k)}$ and $b_{u,t}^{(k)}$ denotes the t-th behavior in the k-th sequence. Given a user $u \in \mathcal{U}$, a candidate item $v \in \mathcal{I}$, and $\mathcal{S}_u$, the ranking model predicts the interaction probability as

$$\hat{y} = \mathbb{P}_\theta\left(y = 1 \mid u, v, \mathcal{S}_u\right), \tag{1}$$

where $\theta \in \Theta$ denotes the model parameters and $y \in \{0, 1\}$ indicates whether user u interacts with item v.

Given a historical dataset $\mathcal{D} = \{(u, v, \mathcal{S}_u, y)\}$, the optimal model parameters are given by

$$\theta^\star = \arg\min_{\theta \in \Theta} \mathcal{L}(\theta), \tag{2}$$

where $\mathcal{L}(\theta)$ is the binary cross-entropy loss:

$$\mathcal{L}(\theta) \triangleq -\frac{1}{|\mathcal{D}|} \sum_{(u,v,\mathcal{S}_u,y)\in\mathcal{D}} \left[y \log \hat{y} + (1-y)\log(1-\hat{y})\right]. \tag{3}$$

3.2 Overall Framework

The overall framework of SPIRE is illustrated in Figure 1. For each historical behavior, SPIRE partitions its features into multiple semantic groups, each designated as shared or private according to transferability, and then projects each group into a semantic token. Shared tokens capture transferable information across sequences, while private tokens preserve sequence-specific characteristics. In parallel, query and selected non-sequential features are similarly tokenized into query and auxiliary tokens, respectively.

Each SPIRE block then performs structured interactions over these tokens through three modules. **Token Encoding** models intra-behavior and inter-behavior dependencies via row-wise and column-wise token attention, respectively. During column-wise interaction, shared tokens exchange information across sequences, while private tokens are restricted to their corresponding sequences. To reduce the computational and memory overhead of column-wise attention, we introduce a sequence compression strategy that uses a short subsequence as queries over the full history. **Target Attention** then uses query tokens to retrieve relevant interests from the encoded behavior histories, producing target-aware representations. Finally, **Token Enhancing** enables interactions between target-aware tokens and auxiliary tokens derived from non-sequential features, followed by token-wise FFNs. This combination helps adapt the learned interests to the prediction context.

Taken together, the shared-private tokenization and structured token interactions enable fine-grained cross-sequence knowledge sharing while preserving sequence-specific preferences.

3.3 Token Generation

For each historical behavior, its features are partitioned into N predefined semantic groups, including N_s shared groups and N_p private groups, where $N = N_s + N_p$. Shared groups capture transferable semantics across sequences, while private groups preserve sequence-specific characteristics. Each semantic group is independently projected into a token using a group-specific feed-forward network (FFN).

Specifically, for the t-th behavior in the k-th sequence, let $\mathbf{G}_{t,i}^{s,(k)}$ denote the i-th shared feature group and $\mathbf{G}_{t,j}^{p,(k)}$ denote the j-th